

Moving particle

Paul Haskins

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$$\vec{E}(x, t) = \frac{q (\gamma (x + \beta ct), \gamma y, \gamma z)^T}{[(\gamma x + \gamma \beta ct)^2 + y^2 + z^2]^{\frac{3}{2}}} \quad (1)$$

$$(2)$$

Electrostatic field of point charge boosted in the x direction and compare to the velocity field in the retarded time formalism

$$\vec{E}(\vec{x}, t) = \left[\frac{q (\vec{n} - \vec{\beta})}{R^2 \gamma^2 (1 - \vec{\beta} \cdot \vec{n})^3} \right]_{kt} \quad (3)$$

Trajectory of a point charge

$$\vec{r}(t) = (-\beta ct, 0, 0)^T \quad (4)$$

$$\vec{x} - \vec{r}(t_{ret}) = (x + \beta ct_{ret}, y, z)^T = R \vec{n}, \vec{n}^2 = 1 \quad (5)$$

with

$$c(t - t_{ret}) = R = |\vec{x} - \vec{r}(t_{ret})| \quad (6)$$

$$\implies ct_{ret} = ct - R \quad (7)$$

$$R \vec{n} = \vec{x} - \vec{r}(t_{ret}) = (x + \beta ct - \beta R, y, z)^T \quad (8)$$

$$R^2 \vec{n}^2 = R^2 = (x + \beta ct - \beta R)^2 + y^2 + z^2 \quad (9)$$

Solve the quadratic for R. We get the solutions

$$R_{\pm} = -\gamma \left[\beta (\gamma x + \gamma \beta ct) \mp \sqrt{(\gamma x + \gamma \beta ct)^2 + y^2 + z^2} \right] \quad (10)$$

Where R is the modulus of a vector and therefore positive. We assume $\beta (\gamma x + \gamma \beta ct) \geq 0$. $\implies R_- < 0$ keep $R = R_+$

$$\left[\vec{n} - \vec{\beta} \right]_{ret} = \frac{1}{R} \left[(\vec{x} - \vec{r}(t_{ret})) - R \vec{\beta} \right] \quad (11)$$

$$= \frac{1}{R} [x + \beta ct - \beta R + R \beta, y, z]^T \quad (12)$$

$$= \frac{1}{R} \begin{pmatrix} x + \beta ct \\ y \\ z \end{pmatrix} \quad (13)$$

We have R times the denomination of $\vec{E}_{ret} = \gamma^2 (R - R \vec{\beta} \cdot \vec{n})^3$.

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$$\left[R - R \vec{\beta} \cdot \vec{n} \right]_{ret} = R - [R \beta n]_{ret} = R + \beta (x + \beta ct) - \beta^2 R = R \gamma^{-2} + \beta (x + \beta ct) \quad (14)$$

$$\frac{R_+}{\gamma^2} + \beta (x + \beta ct) = \frac{1}{\gamma^2} [-\gamma \beta (\gamma x + \gamma \beta ct) + \gamma \sqrt{\dots} + \beta (x + \beta ct) \gamma^2] = \frac{\sqrt{\dots}}{\gamma} \quad (15)$$

$$\left[\frac{q (n - \vec{\beta})}{R^2 \gamma^2 (n - \beta \cdot n)^3} \right]_{ret} = \frac{q (x + \beta ct, y, z)^T}{\sqrt{\dots}} \gamma^3 \quad (16)$$

$$= \frac{q (x + \beta ct, y, z)^T}{\gamma^2 (\gamma^2 [x + \beta ct]^2 + y^2 + z^2)^{\frac{3}{2}} \gamma^{-3}} \quad (17)$$

$$= \frac{\gamma q (x + \beta ct, y, z)^T}{(\gamma^2 (x + \beta ct)^2 + y^2 + z^2)^{\frac{3}{2}}} \quad (18)$$

Electromagnetism in matter (SI)

Maxwell equations in a vacuum.

$$\vec{\nabla} \vec{E} = \frac{\rho}{\epsilon_0} \quad (19)$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad (20)$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad (21)$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{j} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad (22)$$

in the case where we have a medium

$$\nabla \cdot \vec{D} = \rho_{free} \quad (23)$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad (24)$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad (25)$$

$$\vec{\nabla} \times \vec{H} = \vec{j}_{free} + \frac{\partial \vec{D}}{\partial t} \quad (26)$$

$$(27)$$

and we have the electric displacement

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} \quad (28)$$

$$\vec{H} = \frac{1}{\mu_0} \vec{B} - \vec{M} \quad (29)$$

auxiliary field (griffiths) or magnetising field.

In linear media.

$$\vec{P} = \epsilon_0 \chi_e \vec{E} \quad (30)$$

$$\vec{M} = \chi_m \vec{H} \quad (31)$$

$$\implies \vec{D} = \epsilon_0 (1 + \chi_e) \vec{E} \quad (32)$$

we define $\epsilon = \epsilon_0 (1 + \chi_e)$

$$\vec{H} (1 + \chi_m) = \frac{1}{\mu_0} \vec{B} \quad (33)$$

$$\vec{H} = \frac{1}{\mu_0 (1 + \chi_m)} \vec{B} = \frac{1}{\mu} \vec{B} \quad (34)$$